MIL-F-18180/2A
30 May 1986
SUPERSEDING
MIL-F-18180/2
30 April 1981

MILITARY SPECIFICATION SHEET

FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500; SOCKET-WELDING FLANGES

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the socket-welding flanges described herein shall consist of this document and the latest issue of specification MIL-F-18180.



ANSI B16.5 CONVOLUTED

FIGURE 1. Socket-welding flanges.

1. Military specification sheet part number. Socket-welding flanges covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-18180).

M18180/2	XX	X	XX	X
Military Specification Sheet Code Number				
Pipe Size and Pressure-Temperature Rating Class Code Number (see para 2)				
Flange Facing Code Letter (see para 3)				
ANSI B16.5 Materials Group No. Code Number (see para 4)				
Pipe Schedule Code Letter (Waterway Bore for Class 600 or 1500 Only - see para 6)				

FSC 4730

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2. Pipe size and pressure-temperature rating. The nominal pipe size and pressure-temperature rating class of socket-welding flanges are designated by two digits (see table I).

TABLE I. Code number to size and class.

Nominal Pipe Size (Inch)	Pressure-Temperature Ratings			
	Class 150	Class 300	Class 600	Class 1500
1/2	01	21	52	84
3/4	02	22	53	85
1	03	23	54	86
1-1/4	04	24	55	87
1-1/2	05	25	56	88
2	06	26	57	89
2-1/2	07	27	58	90
3	08	28	59	--

3. Flange facing. The facing of socket-welding flanges is designated by a single letter (see table II).

TABLE II. Code letter to flange facing.

Code Letter	Flange facing
A	Raised
B	Flat
C	Ring-Joint
D	Large Male
E	Large Female
F	Small Male
G	Small Female
H	Large Tongue
J	Small Tongue
K	Large Groove
M	Small Groove

4. ANSI B16.5 material group number. The ANSI B16.5 Material Group Number for socket-welding flanges is designated by two digits (see table III).

TABLE III. Code number to material group number.

Code No.	ANSI B16.5 Material Group No.	ANSI B16.5 Acceptable Material (ASTM Standards for)
01	Group 1.1	A105, A216 Grade WCB, A350 Grade LF2, A515 Grade 70, A516 Grade 70 or A537 Class 1.
02	Group 1.2	A203 Class B or E, A216 Grade WCC, A350 Grade LF3 or A352 Grade LC2 or LC3.
04	Group 1.4	A350 Grade LF1, A515 Grade 60 or A516 Grade 60.

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TABLE III. Code number to material group number. - Continued

Code No.	ANSI B16.5 Material Group No.	ANSI B16.5 Acceptable Material (ASTM Standards for)
05	Group 1.5	A182 Grade F1, A204 Grade A or B, A217 Grade WC1 or A352 Grade LC1.
07	Group 1.7	A182 Grade F2, A204 Grade C or A217 Grade WC4 or WC5.
09	Group 1.9	A182 Grade F11 or F12, A217 Grade WC6 or A387 Grade 11.
10	Group 1.10	A182 Grade F22, A217 Grade WC9 or A387 Grade 22.
13	Group 1.13	A182 Grade F5a or A217 Grade C5.
14	Group 1.14	A182 Grade F9 or A217 Grade C12.
21	Group 2.1	A182 Grade F304 or F304H, A240 Type 304 or A351 Grade CF3 or CF8.
22	Group 2.2	A182 Grade F316 or F316H, A240 Type 316 or 317 or A351 Grade CF3M or CF8M.
23	Group 2.3	A182 Grade F304L or F316L or A240 Type 304L or 316L.
24	Group 2.4	A182 Grade F321 or F321H or A240 Type 321 or 321H.
25	Group 2.5	A182 Grade F347, F347H, F348 or F348H, A240 Type 347, 347H, 348 or 348H, or A351 Grade CF8C.
26	Group 2.6	A240 Type 309S or A351 Grade CH8 or CH20.
27	Group 2.7	A182 Grade F310, A240 Type 310S or A351 Grade CK20.

5. Waterway bore diameter. The waterway bore diameter for socket-welding flanges of classes 150 and 300 shall be as specified in ANSI B16.5. Classes 600 and 1500 socket-welding flanges shall have the waterway bore diameter equal to inside diameter of pipe for pipe schedule (see para 6) and nominal pipe size (see para 2) specified.

6. Pipe schedule. The pipe schedule which is used to specify the waterway bore diameter (inside diameter of pipe for schedule specified) of classes 600 and 1500 socket-welding flanges is designated by a single letter (see table IV).

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TABLE IV. Code letter to pipe schedule.

Code Letter	Pipe Schedule
A	Schedule 40 and Standard Wall
D	Scheule 80 and Extra Strong Wall
H	Schedule 160
J	Double Extra Strong Wall

7. Military specification part number definition. The military specification sheet part number is a definitive part number which is formed by combining the military specification sheet code number (M18180/2 for MIL-F-18180/2) the pipe size and pressure-temperature rating class code number, the flange facing code letter, the ANSI B16.5 materials group number code number and, if applicable, the pipe schedule code letter as shown in paragraph 1.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-18180/2A	FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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